INMO 2013

Exercise 1 (INMO 2013 P4, , Putnam 2002 A3)**.** [**AF13**, Exercise 4.8, p. 90] Let n be an integer greater than 1 and let T_n be the number of nonempty subsets S of $\{1, 2, \ldots, n\}$ with the property that the average of the elements of S is an integer. Prove that $T_n - n$ is always even.

> **Walkthrough** —
>
> **(a)** Show that the average of any nonempty subset of $\{1, 2, \ldots, n\}$ lies between 1 and n.
>
> **(b)** Observe that it suffices to show that for any $1 \leq a \leq n$, the subsets of $\{1, 2, \ldots, n\}$ with average equal to a is an odd number.
>
> **(c)** Show that for any $1 \leq a \leq n$, the nonempty subsets of $\{1, 2, \ldots, n\}$ having average equal to a and not containing a, are in one-to-one correspondence with the subsets of $\{1, 2, \ldots, n\}$ having average equal to a, and which contain a. (Consider the averages of $\{1, 2, 3\}$ and $\{2\}$.)

Solution 1. Let $\mathcal{T}_n$ denote the set of nonempty subsets of $\{1, 2, \ldots, n\}$ having integer average. Note that the average of the elements of any subset of $\{1, 2, \ldots, n\}$ lies between 1 and n. So if a nonempty subset S of $\{1, 2, \ldots, n\}$ has integer average a and S does not contain a, then $S \cup \{a\}$ is also a nonempty subset of $\{1, 2, \ldots, n\}$ with average a. Hence, for any integer $1 \leq a \leq n$, there is a bijection between the nonempty subsets of X with integer average a that does not contain a and the subsets of X with integer average a that contain a, but not equal to $\{a\}$. So the subsets of $\mathcal{T}_n$ with average equal to a is an odd number for any $1 \leq a \leq n$. Hence $T_n - n$ is even. ■

References

[**AF13**] T. Andreescu and Z. Feng. *A Path to Combinatorics for Undergraduates: Counting Strategies*. Birkhäuser Boston, 2013. ISBN: 9780817681548. URL: https://books.google.de/books?id=3mwQBwAAQBAJ (cited p. 1)